

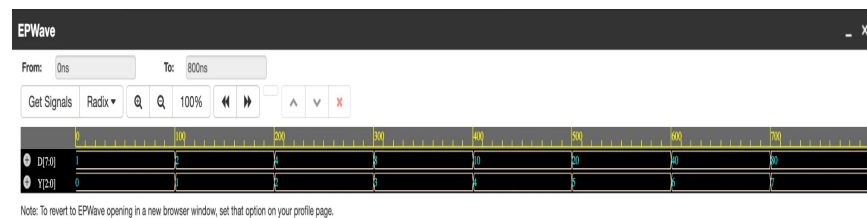
Experiment 5 Assignment

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2025A7PS0443U
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Q1: 8-to-3 Priority Encoder

<https://edaplayground.com/x/m3iX>

Code:



Waveform:

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Notes: Does not endorse testing material here. Not suitable for EDA Playground.

Languages & Libraries

Tools & Simulators

Cadence Xcelium 25.03

Compile Options

Immscale 1ns/1ns -yysv

Run Options

access +rw

Use run.do Tcl file

Use run.bash shell script

Open EPWave after run

Show output text file

Show output SVG file

Download files after run

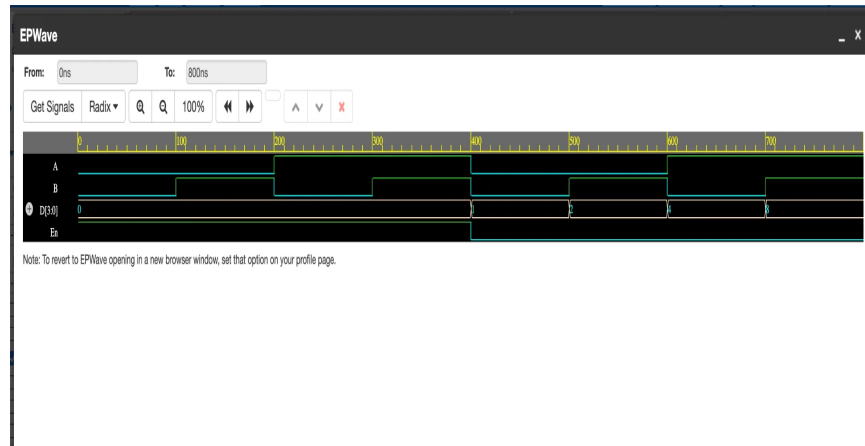
```
1 module tb_encoder;
2 reg [7:0] D;
3 wire [2:0] Y;
4 initial
5 begin
6   $dumpfile ("dump.vcd");
7   $dumpvars (1, tb_encoder);
8   #0 D <= 8'b00000001;
9   #100 D <= 8'b00000010;
10  #100 D <= 8'b00000100;
11  #100 D <= 8'b00001000;
12  #100 D <= 8'b00010000;
13  #100 D <= 8'b00100000;
14  #100 D <= 8'b01000000;
15  #100 D <= 8'b10000000;
16  #100 $stop;
17 end
18 encoder_8to3 U1(Y,D);
19 endmodule
```

```
1 module encoder_8to3 (Y,D);
2 input [7:0] D;
3 output reg [2:0] Y;
4 always @(D)
5 begin
6   case (D)
7     8'b00000001: Y = 3'b000;
8     8'b00000010: Y = 3'b001;
9     8'b00000100: Y = 3'b010;
10    8'b00001000: Y = 3'b011;
11    8'b00010000: Y = 3'b100;
12    8'b00100000: Y = 3'b101;
13    8'b01000000: Y = 3'b110;
14    8'b10000000: Y = 3'b111;
15    default: Y = 3'bxxx;
16  endcase
17 end
18 endmodule
```

Q2: 2-to-4 Decoder with Active-Low Enable

<https://edaplayground.com/x/Nunh>

Code:



Waveform:

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• Languages & Libraries

• Tools & Simulators

Cadence Xcelium 25.03

Compile Options

-timescale 1ns/1ns -sysv

Run Options

-access +rw

☐ Use run.do Tcl file

☐ Use run.bash shell script

☒ Open EPWave after run

☐ Show output text file

☐ Show output SVG file

☐ Download files after run

testbench.sv

```
1 module tb_decoder;
2   reg A,B,En;
3   wire [3:0] D;
4   initial
5   begin
6     $dumpfile ("dump.vcd");
7     $dumpvars (1, tb_decoder);
8     #000 En=1; A=0; B=0;
9     #100 En=1; A=0; B=1;
10    #100 En=1; A=1; B=0;
11    #100 En=1; A=1; B=1;
12    #100 En=0; A=0; B=0;
13    #100 En=0; A=0; B=1;
14    #100 En=0; A=1; B=0;
15    #100 En=0; A=1; B=1;
16    #100 $stop;
17  end
18  decoder_2to4 U1(D,A,B,En);
19 endmodule
```

design.sv

```
1 module decoder_2to4 (D,A,B,En);
2   input A,B,En;
3   output [3:0] D;
4   assign D[0] = (~A)&B&~En;
5   assign D[1] = (~A)&B&En;
6   assign D[2] = (A)&B&~En;
7   assign D[3] = (A)&B&En;
8 endmodule
```

Q3: 4-bit Magnitude Comparator

<https://edaplayground.com/x/UjD5>

Code:

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Languages & Libraries

Tools & Simulators

Cadence Xcelium 25.03

Compile Options

timescale 1ns/1ns -sysv

Run Options

access +rw

Use run.do To file

Use run.bash shell script

Open EPWave after run

Show output text file

Show output SVG file

Download files after run

testbench.v

```
1 module tb_comparator;
2 reg [3:0] A,B;
3 wire L,G,E;
4 initial
5 begin
6 $dumpfile ("dump.vcd");
7 $dumpvars (1, tb_comparator);
8 #000 A=4'b0000; B=4'b1111;
9 #100 A=4'b1111; B=4'b0000;
10 #100 A=4'b0000; B=4'b0000;
11 #100 A=4'b1111; B=4'b1111;
12 #100 A=4'b0101; B=4'b1000;
13 #100 A=4'b1010; B=4'b0011;
14 #100 A=4'b1001; B=4'b1001;
15 #100 $stop;
16 end
17 comparator_4bit U1(L,G,E,A,B);
18 endmodule
```

design.v

```
1 module comparator_4bit (L,G,E,A,B);
2 input [3:0] A,B;
3 output reg L,G,E;
4 always @(A or B)
5 begin
6 if (A > B)
7 begin
8 G = 1; E = 0; L = 0;
9 end
10 else if (A == B)
11 begin
12 G = 0; E = 1; L = 0;
13 end
14 else
15 begin
16 G = 0; E = 0; L = 1;
17 end
18 end
19 endmodule
```

Waveform:

